Sensorless Control of Ultrahigh-Speed PM Brushless Motor Using PLL and Third Harmonic Back EMF

J. X. Shen, *Senior Member, IEEE*, and S. Iwasaki

***Abstract*—Application-specific integrated circuit (ASIC) ML4425 is often used for sensorless control of permanent-magnet (PM) brushless direct current (BLDC) motor drives. It integrates the terminal voltage of the unenergized winding that contains the back electromotive force (EMF) information and uses a phase-locked loop (PLL) to determine the proper commutation sequence for the BLDC motor. However, even without pulsewidth modulation, the terminal voltage is distorted by voltage pulses due to the freewheel diode conduction. The pulses, which appear very wide in an ultrahigh-speed (120 kr/min) drive, are also integrated by the ASIC. Consequently, the motor commutation is significantly retarded, and the drive performance is deteriorated. In this paper, it is proposed that the ASIC should integrate the third harmonic back EMF instead of the terminal voltage, such that the commutation retarding is largely reduced and the motor performance is improved. Basic principle and implementation of the new ASIC-based sensorless controller will be presented, and experimental results will be given to verify the control strategy. On the other hand, phase delay in the motor currents arises due to the influence of winding inductance, reducing the drive performance. Therefore, a novel circuit with discrete components is proposed. It also uses the integration of third harmonic back EMF and the PLL technique and provides controllable advanced commutation to the BLDC motor.**

***Index Terms*—Application-specific integrated circuit (ASIC), brushless direct current (BLDC) motor, phase-locked loop (PLL), sensorless control, third harmonic back electromotive force (EMF), ultrahigh-speed motor.**

I. Introduction

SENSORLESS control is often required in permanent-magnet (PM) brushless direct current (BLDC) motor drives. There are many categories of sensorless control strategies; however, the most popular category is based on back electromotive forces (EMFs). Basically, the EMFs can be utilized in three ways: 1) detecting the EMF zero crossings; 2) integrating the EMF from its zero crossing and comparing the integration result with a reference voltage; and 3) integrating the EMF and using a phase-locked loop (PLL) to keep the integration result being zero. On the other hand, either phase EMF or third harmonic EMF can be used for sensorless control, whereas the phase EMF is usually extracted from the terminal voltage of the unenergized phase winding. Therefore, there should be six EMF-based sensorless control techniques, as classified in Table I. Among them, techniques I–V have been reported, major references being given in Table I, whereas technique VI is novel and will be presented in this paper.

TABLE I
CLASSIFICATION OF BACK EMF-BASED SENSORLESS CONTROL TECHNIQUES AND MAJOR REFERENCES

utilization / signal	zero-crossing detection		integration & comparing		integration & PLL	
phase EMF	I	[1],[2]	II	[3],[4]	III	[5]-[7]
3rd harmonic	IV	[8]-[10]	V	[11]	VI	this paper

TABLE II
SPECIFICATION OF EXPERIMENTAL BLDC MOTOR

Rated speed (kr/min)	120
Rated DC voltage (V)	288
Rated power (kW)	5
Number of phases	3
Number of poles	4
Number of slots	12
Self-inductance (mH)	0.12
Mutual-inductance (mH)	-0.05
Resistance @ 30°C (mΩ)	31.5

ML4425, a commercial application-specific integrated circuit (ASIC), which is based on the abovementioned technique III [6], [7], is often used for sensorless control of BLDC drives. It integrates the terminal voltage of the unenergized winding that contains the phase EMF information, and its PLL ensures that the integration result be zero. Thus, the BLDC motor can be commutated with a proper frequency and phase angle. In most cases, the ASIC provides very reliable operation, and its peripheral circuit is simple. Therefore, the ASIC is used in an experimental ultrahigh-speed (120 kr/min) system. The motor specification is listed in Table II. Usually, the major problem that the ASIC encounters is the inferior starting performance, as open-loop starting procedure is applied. However, in the experimental system, the BLDC motor drives an air compressor that requires low starting torque. Therefore, the open-loop starting can be easily realized. However, it is found that the commutation of the BLDC drive is significantly retarded during high-speed operation. This is because a relatively wide voltage pulse due to the freewheel diode conduction appears in the terminal voltage of the unenergized winding and is also integrated by the ASIC. Further analysis will be presented hereafter. Consequently, the commutation retarding causes low power factor, low efficiency, low output power capability, and

Manuscript received July 2, 2004; revised October 12, 2004. Abstract published on the Internet January 25, 2006. An early version of this paper was presented at the 2003 IEEE International Electric Machines and Drives Conference, Madison, WI, June 1–4.

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Digital Object Identifier 10.1109/TIE.2006.870664

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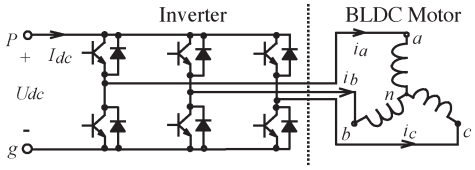


Fig. 1. Schematic diagram of BLDC drive.

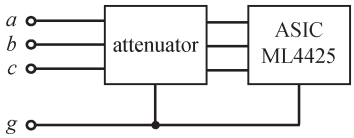


Fig. 2. Signal connection in original controller.

high temperature rise. To overcome the problem, it is proposed that the ASIC should integrate the third harmonic back EMF instead of the terminal voltage. Some preliminary results of this technique were reported in an early paper [12], whereas further study and improvement will be presented hereafter.

It should be pointed out that third harmonic back EMF has been used in sensorless controls by detecting the zero crossings with a voltage comparator [9], [10] or by integrating the EMF and comparing with a reference voltage [11]. Those methods, which are sensitive to the noise in the third harmonic EMF signal, belong to the abovementioned techniques IV and V, respectively. However, the sensorless technique presented in this paper uses a voltage integrator and a PLL to process the third harmonic EMF. Hence, it is robust to the signal noise and belongs to technique VI.

II. Original Sensorless Control

Fig. 1 shows the schematic of a BLDC drive. The motor terminal voltages, referred to as $u_{\rm ag}$, $u_{\rm bg}$, and $u_{\rm cg}$, are used by the ASIC for the sensorless control, as shown in Fig. 2. From the terminal voltages, the ASIC creates a virtual neutral with its internal circuit. On the other hand, if the freewheel diode conduction is not considered, at any time, there must be one phase that is not energized, and each unenergizing status lasts 60 electrical degrees. Hence, from the virtual neutral and the terminal voltage of the unenergized phase, the back EMF is obtained. The ASIC integrates the back EMF and at the same time generates the motor commutation sequence in such a way that the integration result, i.e., the net shaded area in Fig. 3, is zero. Consequently, each phase must stop energizing at 30 electrical degrees prior to the occurrence of back EMF zero crossing and start energizing oppositely at 30 electrical degrees after the same zero crossing, as shown in Fig. 3. Hence, a PLL is formed with the controller, inverter, and motor, making the motor commutation in phase with the back EMF.

However, the terminal voltage of the unenergized phase actually contains pulses due to the freewheel diode conduction, as shown in Fig. 4(a). The pulses are also integrated by the ASIC. Consequently, when the integration result, i.e., the net shaded area in Fig. 4(a), is zero, the motor commutation drifts away from the desired phase angle, causing a commutation advancing. Furthermore, the drifting angle varies with the motor parameters, speed and load condition; hence, it is out of control.

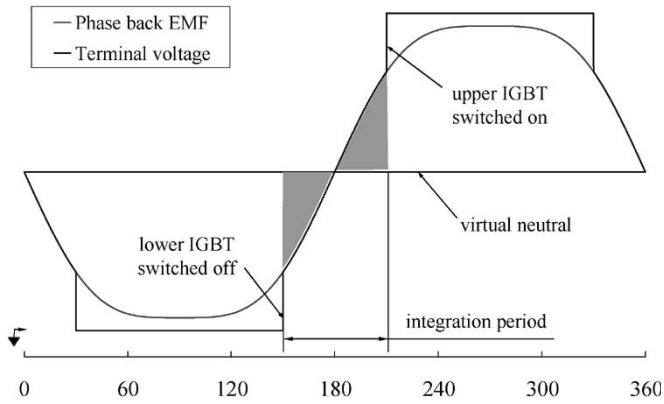


Fig. 3. Principle of original controller without considering freewheel diode conduction.

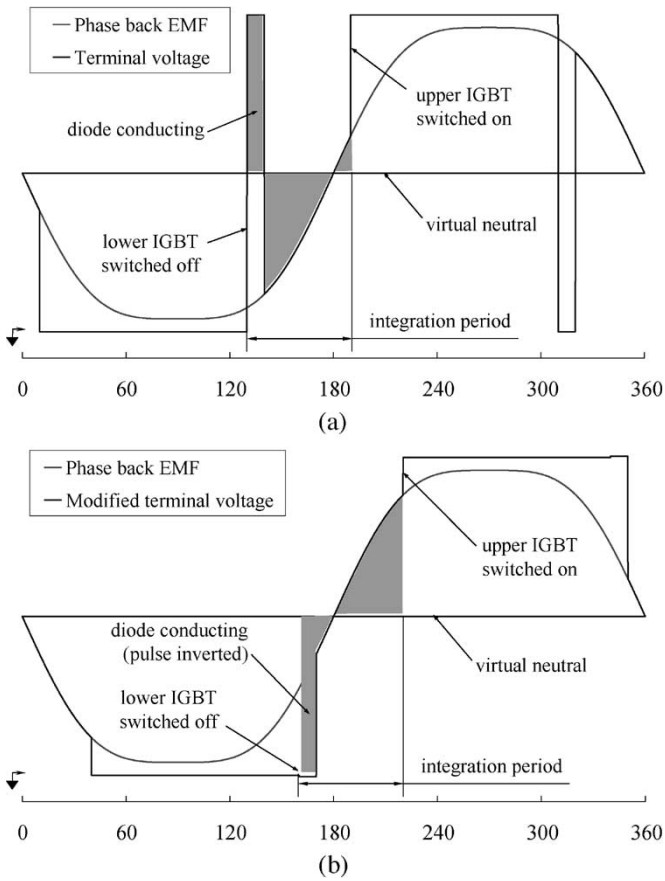


Fig. 4. Operation of original controller with freewheel diode conduction considered. (a) Voltage pulse not inverted. (b) Voltage pulse inverted.

In Fig. 4(a), it is noticed that the voltage pulse is far away from the back EMF waveform. For example, the shaded pulse is positive, whereas the back EMF at the same time is negative. Thus, the pulse distorts the EMF waveform significantly. However, if the voltage pulse is inverted, it will be closer to the back EMF, as shown in Fig. 4(b), both being negative in the example. Hence, the EMF waveform is less distorted, and the phase drifting angle will be smaller. However, the commutation will become retarded, and the retarding angle is also out of

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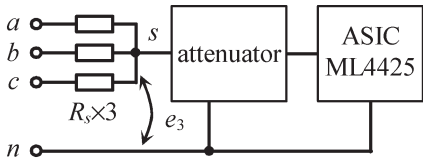


Fig. 5. Signal extraction and connection in improved controller.

control. This technique will actually be used in the ASIC-based sensorless controller.

The voltage pulse will be remarkable if the motor electric time constant is big or the motor is heavily loaded, as it takes long time for the current of the unenergized phase to decay through the freewheel diode. Furthermore, if the motor speed is high, the fundamental cycle is short; hence, the pulse appears relatively wide. Particularly, in the experimental system, because the BLDC motor drives an air compressor, the higher the motor speed is, the heavier the load. Therefore, the commutation retarding will be rather serious during ultrahigh-speed operation. This will be observed from experimental results in Section IV.

III. Improved Sensorless Control

The commutation retarding under the original control is caused by the waveform distortion in the sensed signals. To solve this problem, the signal that contains the EMF information but is not distorted, such as the third harmonic back EMF, should be used for the ASIC control. The phase EMF of most BLDC motors is close to a trapezoidal waveform; hence, it contains third harmonic component, as expressed below

$$\begin{cases} e_a = e_{a1} + e_3 + e_{a5} + e_{a7} + e_9 + \cdots \\ e_b = e_{b1} + e_3 + e_{b5} + e_{b7} + e_9 + \cdots \\ e_c = e_{c1} + e_3 + e_{c5} + e_{c7} + e_9 + \cdots \\ e_{a1} = K_{E1}\omega \sin\theta \\ e_{b1} = K_{E1}\omega \sin\left(\theta - \frac{2\pi}{3}\right) \\ e_{c1} = K_{E1}\omega \sin\left(\theta + \frac{2\pi}{3}\right) \\ e_3 = K_{E3}\omega \sin 3\theta \\ \cdots \\ \sum\limits_{a,b,c} e_{k1} = \sum\limits_{a,b,c} e_{k5} = \sum\limits_{a,b,c} e_{k7} = \cdots = 0 \end{cases} \tag{1}$$

where the number in the subscript denotes the harmonic order, k denotes phases a, b, or c, K_E represents the EMF constant of each harmonic, θ is the rotor position, and ω is the electric angular speed. It is seen that the third harmonic EMF e_3 is a function of θ; hence, it contains the rotor position information. Furthermore, e_3 can be easily extracted from the motor winding terminals a, b, and c and the neutral point n with a simple Y-connected resistor network [8]–[10], as shown in Fig. 5. From the resistor network, the following voltage equation is obtained:

$$\begin{aligned} u_{an} + u_{bn} + u_{cn} &= u_{as} + u_{bs} + u_{cs} + 3u_{sn} \\ &= (i_{as} + i_{bs} + i_{cs})R_s + 3u_{sn} \\ &= 3u_{sn} \end{aligned} \tag{2}$$

where i_{as}, i_{bs}, and i_{cs} are the currents in the resistors, and $i_{as} + i_{bs} + i_{cs} = 0$ if the attenuator input impedance is very high. Moreover, assuming constant winding inductance L and M in a BLDC motor with surface-mounted PMs (SPMs), the motor phase voltage and current equations are

$$\begin{cases} \begin{bmatrix} u_{an} \\ u_{bn} \\ u_{cn} \end{bmatrix} = R \times \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix} + \begin{bmatrix} L & M & M \\ M & L & M \\ M & M & L \end{bmatrix} \\ \qquad\qquad \times \frac{d}{dt} \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix} + \begin{bmatrix} e_a \\ e_b \\ e_c \end{bmatrix} \\ i_a + i_b + i_c = 0 \end{cases} \tag{3}$$

where i_a, i_b, and i_c are the currents in the motor windings. From (1)–(3), the following is derived:

$$u_{sn} = e_3 + e_9 + e_{15} + \cdots \approx e_3. \tag{4}$$

Therefore, the third harmonic EMF can be extracted across the resistor network star point (s) and the motor winding neutral point (n). Moreover, the above equation is always valid in spite of the insulated gate bipolar transistor (IGBT) switching or freewheel diode conducting status; thus, the extracted signal waveform is theoretically undistorted.

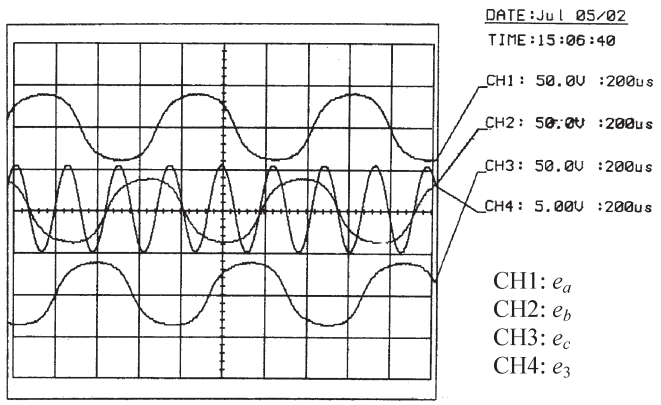


Fig. 6. Waveforms of phase back EMFs and third harmonic back EMF in experimental BLDC motor.

Fig. 6 shows the captured waveforms of three-phase and third harmonic back EMFs in the experimental motor. It is seen that the zero crossings of phase EMFs occur at the same time as those of the third harmonic EMF. On the other hand, during BLDC operation, the phase EMF cannot be captured directly because the inverter applies voltage on the winding terminals. However, if the third harmonic EMF is measured from the signal u_{sn}, the phase EMF can then be synthesized from its phase angle and harmonic contents that are measured in advance with a signal analyzer.

Fig. 7 shows the principle of the improved sensorless controller. In the steady state, the integration result of the third harmonic EMF, i.e., the net shaded area, is zero. Thus, an inverter IGBT is switched off at 30 electrical degrees prior to the occurrence of the back EMF zero crossing, and another IGBT of the same phase is switched on at 30 electrical degrees after the same zero crossing. Hence, normal commutation is realized. However, during the transient state, if the motor commutation is advanced, the integration result will be negative. This

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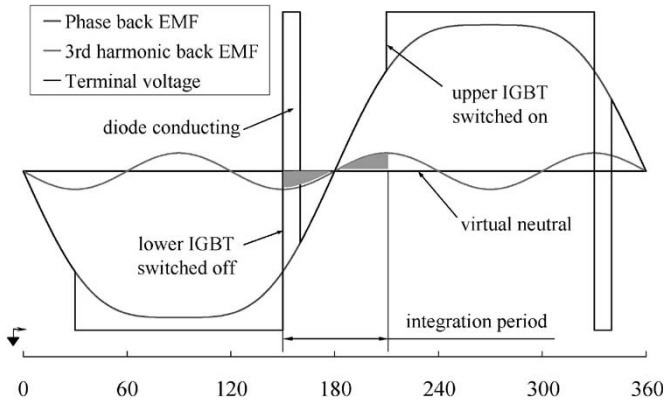


Fig. 7. Principle of improved controller.

will reduce the input voltage of a voltage-controlled oscillator (VCO) in the PLL and consequently decrease the motor commutation frequency until the commutation is no longer advanced. In contrast, if the commutation is retarded, the integration result will be positive; hence, the VCO input and motor commutation frequency will be increased until the commutation is no longer retarded. Furthermore, it is seen from Fig. 7 that the freewheel diode conduction does not present a voltage pulse in the signal integrated by the ASIC; thus, during the steady state, it would not cause retarding or advancing in the motor commutation.

IV. Experimental Investigation

Both original and improved sensorless controllers are investigated with the same IGBT intelligent power module (IPM) inverter, BLDC motor, and air compressor load. The third harmonic back EMF is always captured, and the phase back EMF is then synthesized. The phase current waveform is also captured, from which the motor commutation can be recognized. If the normal commutation is achieved, the inverter switching status should change at the instant when the third harmonic EMF reaches its peak value, as can be seen from Fig. 7.

When the experimental BLDC motor is operated with the original sensorless controller at low speed (10 kr/min), the current is low, and the width of pulse in the terminal voltage is very short. Hence, the motor commutation is hardly retarded, as evident in Fig. 8(a). However, if the motor runs at ultrahigh speed (120 kr/min), the voltage pulse is rather wide. Hence, the commutation is retarded by 12.9 electrical degrees in the test. Furthermore, the retarded commutation, in turn, makes the current decaying period longer, being 35.2 electrical degrees in Fig. 8(b).

On the other hand, if the BLDC motor is operated with the improved controller, the commutation at low speed (10 kr/min) is not retarded either, as indicated in Fig. 9(a). However, at ultrahigh speed (120 kr/min), although the freewheel diode conducting period is long, the commutation retarding is significantly reduced to 3.6 electrical degrees in the test. Such a retarding is mainly caused by some low-pass filters used in the controller. Furthermore, with less commutation retarding, current through the freewheel diode decays faster by 21.6 electrical degrees in Fig. 9(b).

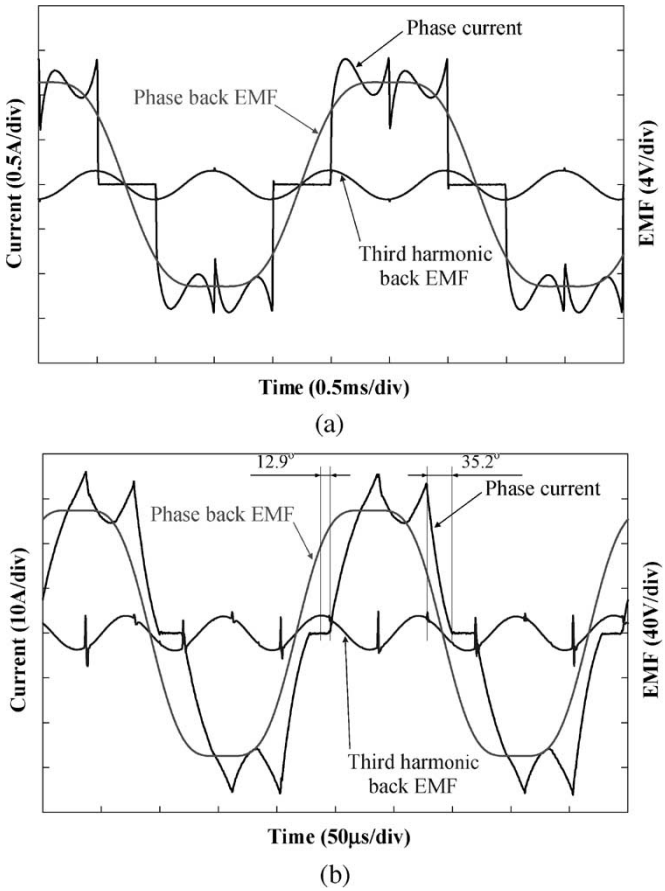


Fig. 8. Waveforms of phase current (captured), third harmonic back EMF (captured), and phase back EMF (synthesized) when BLDC motor is operated with original sensorless controller. (a) 10 kr/min, 21.7 V dc, 1.18 A dc. (b) 120 kr/min, 272.7 V dc, 21.6 A dc.

The above experimental results prove that the original controller exhibits inferior performance when the freewheel diode conduction is long, whereas the improved controller is not affected by the diode conduction and, hence, is suitable for the full speed range and all load conditions. This is further verified with the experimental comparison of the two controllers, where the motor operation conditions are the same. The dc and 3ϕ voltages, currents, and powers, as well as the motor power factor and speed, are measured with a Yokogawa PZ4000 power analyzer. Furthermore, thermocouples are embedded in the motor slots and end windings. It is found that the end-winding temperature is higher than the slot-winding temperature; hence, it is recorded for comparison. On the other hand, the average temperature of rotor magnets are estimated as below. First, the magnet temperature coefficient is premeasured. Second, the motor open-circuit EMF constant is measured at a known room temperature. Finally, when the motor is operated under a certain load condition, the controller is suddenly shut down, and the motor windings are disconnected from the inverter, and the open-circuit EMF constant is remeasured immediately. Thus, from the magnet temperature coefficient as well as the variation of EMF constant, the on-load magnet temperature is calculated. Both winding and magnet temperatures are measured in the thermal steady state. For example, Fig. 10(a) shows that with the improved controller, the commutation is hardly retarded

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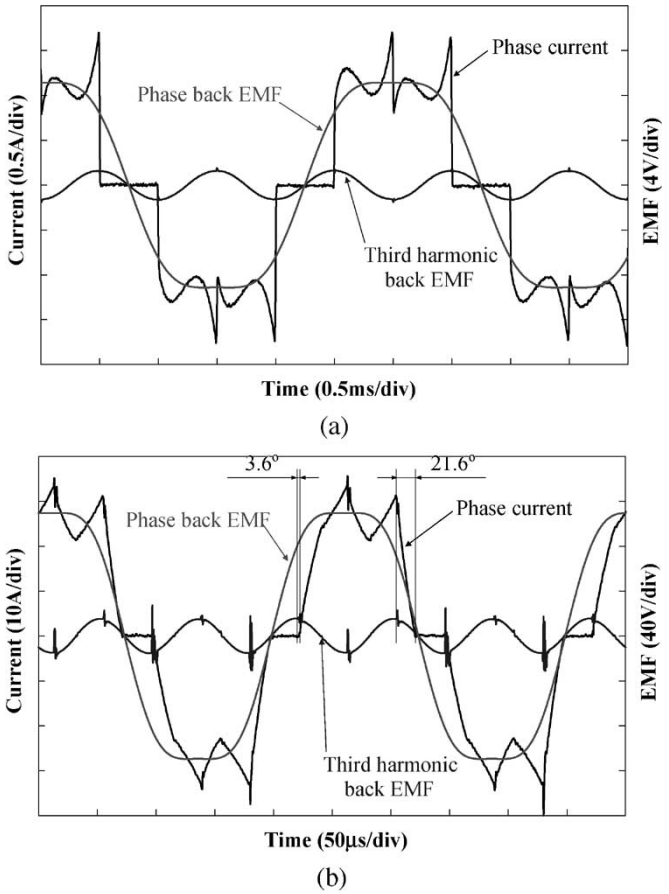


Fig. 9. Waveforms of phase current (captured), third harmonic back EMF (captured), and phase back EMF (synthesized) when BLDC motor is operated with improved sensorless controller. (a) 10 kr/min, 19.5 V dc, 1.14 A dc. (b) 120 kr/min, 261.7 V dc, 21.6 A dc.

and the freewheel diode conduction becomes shorter. Thus, the phase current is less delayed from the phase back EMF, and the equivalent power factor is improved [Fig. 10(b)]. Consequently, the BLDC motor needs less current to achieve the same speed and load [Fig. 10(c)], generates less losses [Fig. 10(d)], requires less input power [Fig. 10(e)], and causes lower temperature rise [Fig. 10(f)]. It should be pointed out that the reduction of temperature rise is crucial in the experimental system, as at rated speed, both permanent magnets and carbon fiber sleeve that is used to protect the magnets are close to their maximum operation temperature. By using the improved sensorless controller, the rotor temperature is reduced by ~ 10 °C. This is rather beneficial to improve the motor reliability. In particular, when the original controller was used, the rotor temperature could reach ~ 145 °C, and some sleeves had broken. However, because the improved controller was used, under the same operation condition, such incident has never happened in the last 1.5 years of operation.

V. Discussion

A. Noise in Third Harmonic Back EMF Signal

It is observed from Figs. 8 and 9 that noise also exists in the extracted signal of the third harmonic EMF. However, the noise is insignificant because its width is much shorter than the cycle of the third harmonic EMF. Moreover, the noise appears as an oscillation if its waveform is zoomed in with an oscilloscope, whereas the integration of oscillation is almost equal to zero. Therefore, unlike the influence of the pulse in the terminal voltage on the original controller, the noise in the third harmonic EMF signal does hardly deteriorate the performance of the improved controller.

B. Low-Speed Performance

The amplitude of phase back EMF is usually higher than that of the third harmonic EMF; hence, it might be queried whether the original controller provides better low-speed performance than the proposed controller. However, this is not the case. The maximum input voltage of the ASIC is 12 V, whereas in the experimental BLDC motor, the amplitudes of both terminal voltage and third harmonic back EMF are higher. Therefore, the terminal voltage or the third harmonic EMF must be attenuated before linking to the ASIC, as shown in Figs. 2 and 5, respectively. The amplitude of third harmonic EMF is about one-eighth of the phase EMF; hence, the attenuation in the improved controller is also about one-eighth of that in the original controller. Therefore, at a certain speed, the amplitude of the effective signal that is directly linked to the ASIC is almost the same in both controllers. Therefore, their low-speed performances are similar. This has been verified with the experimental results, as the minimum stable speed is 5.4 kr/min with the original controller and 3.0 kr/min with the improved controller, the difference being minimal if compared with the motor speed range.

C. Operation Without Back EMF Zero Crossings Present in Terminal Voltages

In most cases the freewheel diode conduction lasts less than 30 electrical degrees; hence, the back EMF zero crossings are present in the terminal voltage of the unenergized phase. However, in some cases such as in ultrahigh-speed motors with big electric time constant (L/R) or with heavy load, the freewheel diode conduction may be longer than 30 electrical degrees. In these cases, the zero crossings do not appear in the terminal voltage waveforms [13]. Therefore, in the original sensorless controller, the voltage integration of the unenergized phase can never be zero, resulting in a wrong motor commutation sequence. However, such a problem can be overcome with the improved sensorless controller because the voltage to be integrated is not distorted by the freewheel diode conduction.

D. Phase Delay of Motor Current Due to Winding Inductance

Although the improved sensorless controller makes the motor commutation almost in phase with the phase back EMF, the actual current is still delayed, as it takes time for the current to increase and decay when the inverter IGBT is switched on and off, respectively, due to the influence of winding inductance.

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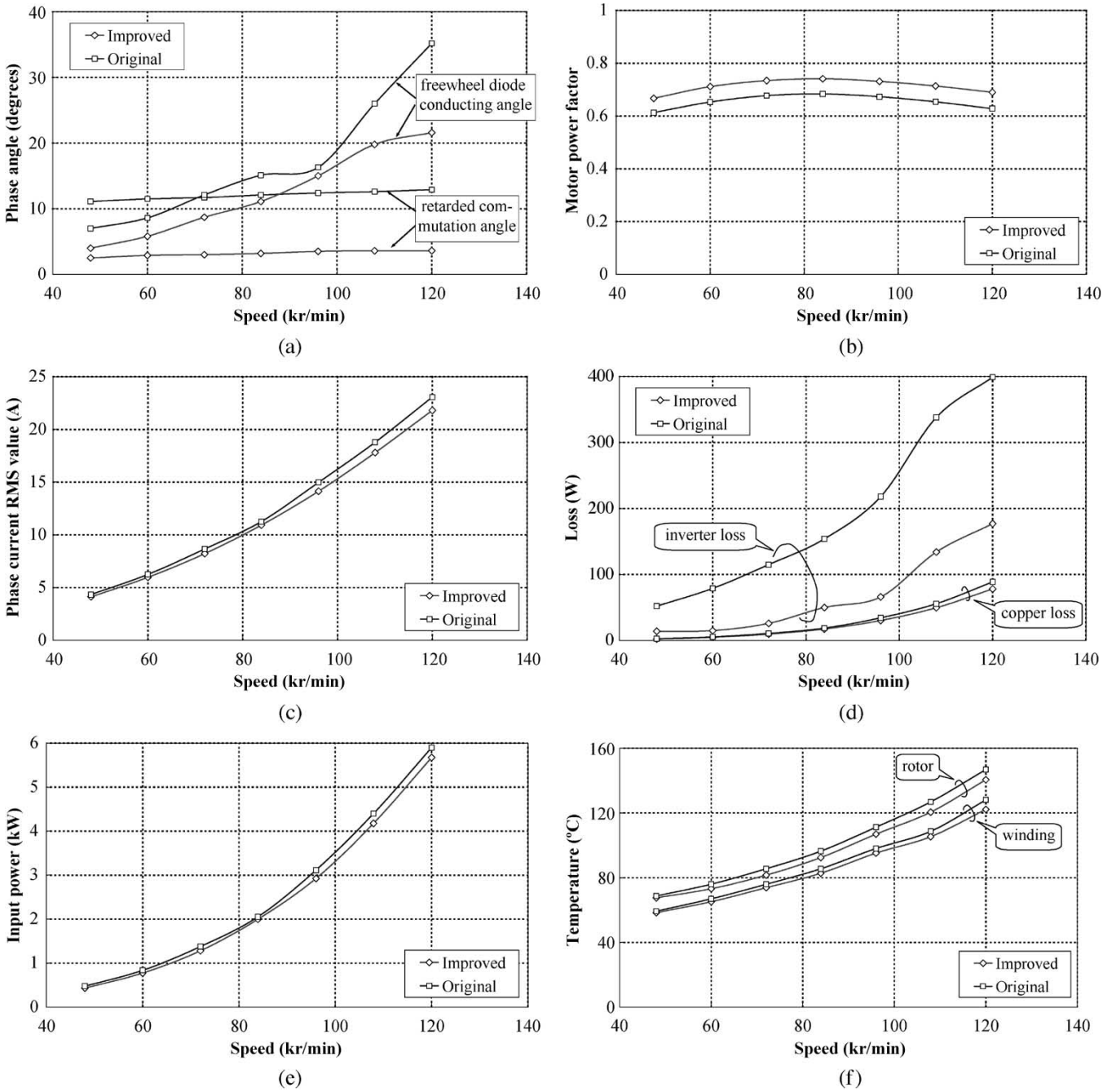


Fig. 10. Experimental comparison of original and improved sensorless controllers over wide speed range. (a) Commutation retarding and diode conducting angle. (b) Motor power factor. (c) Motor phase current root-mean-square (rms) value. (d) Inverter loss and motor copper loss. (e) dc link input power. (f) Rotor magnet and winding temperatures.

For example, the fundamental component of phase current is retarded from the phase EMF by $\sim$ 14 electrical degrees in Fig. 9(b). Such phase delay decreases the motor power factor and causes some further drawbacks. Therefore, a controller that has the feature of advanced commutation is proposed hereafter.

For example, the controller can be made of discrete components, as shown in Fig. 11(a). The circuit inside the dashed-line box has the similar feature to the ASIC ML4425. It preconditions first the third harmonic back EMF e_3, resulting in a new signal e_{3i}, as shown in Fig. 11(b) and (c). Then, it integrates e_{3i}. However, unlike the processing in the ASIC, the integration result V_{3i} is hereby added with a positive voltage reference $V_{\rm ref}$ before connecting to a voltage-controlled oscillator. The digital output signal (VCO) of the oscillator then triggers the inverter. Consequently, the full system, viz., the controller, inverter, and motor, forms a PLL. In the steady state, the PLL ensures that $V_{3i} + V_{\rm ref} = 0$. If $V_{\rm ref}$ is zero, V_{3i}, namely, the shaded area in Fig. 11(b), is also zero; hence, normal commutation is realized. This is the same as the improved ASIC-based sensorless controller. However, if $V_{\rm ref} > 0$, V_{3i} will be negative, as illustrated by the shaded area in Fig. 11(c); thus, advanced commutation is achieved. Moreover, the advanced commutation angle (δ) is

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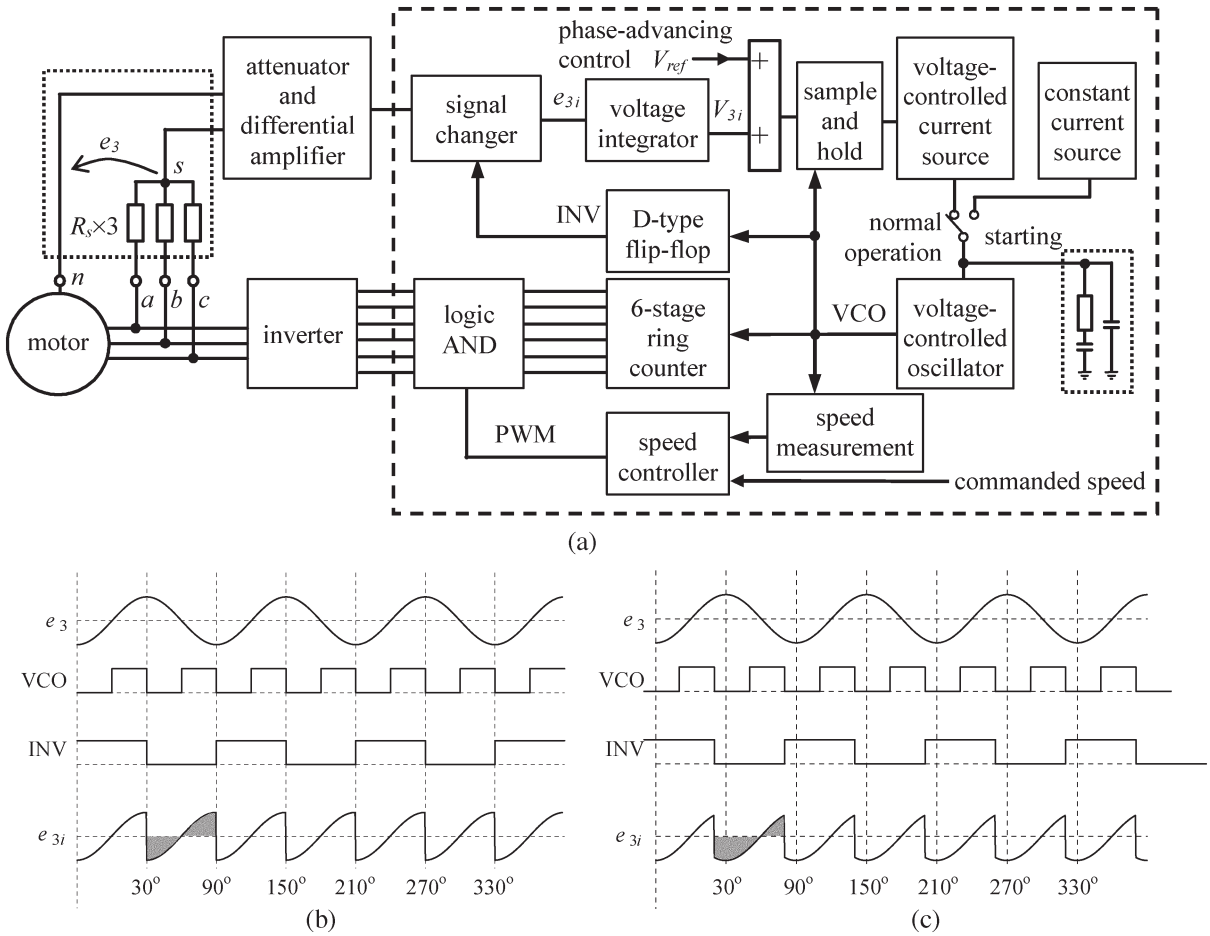


Fig. 11. Principle and waveforms of discrete component-based sensorless controller with feature of advanced commutation. Note: Motor commutation occurs at the instant of falling edge of signal VCO in (b) and (c). (a) Block diagram. (b) Waveforms without commutation advancing when $V_{\rm ref} = 0$. (c) Waveforms with commutation advancing when $V_{\rm ref} > 0$.

controllable. Assuming the third harmonic back EMF constant is K_{E3}, the integration result is then derived as

$$\begin{aligned} V_{3i} &= \int_{\frac{\left(-\frac{\pi}{6}-\delta\right)}{\omega}}^{\frac{\left(\frac{\pi}{6}-\delta\right)}{\omega}} e_{3i}\,dt \\ &= \int_{\frac{\left(-\frac{\pi}{6}-\delta\right)}{\omega}}^{\frac{\left(\frac{\pi}{6}-\delta\right)}{\omega}} [k_a K_{E3}\omega \sin 3\theta]\,dt \\ &= \int_{\frac{\left(-\frac{\pi}{6}-\delta\right)}{\omega}}^{\frac{\left(\frac{\pi}{6}-\delta\right)}{\omega}} [k_a K_{E3}\omega \sin 3\omega t]\,dt \\ &= -\frac{2}{3} k_a K_{E3} \sin 3\delta \end{aligned} \tag{5}$$

where k_a is the attenuation from e_3 to e_{3i}. Therefore, if the voltage reference is set to the following value:

$$V_{\rm ref} = -V_{3i} = \frac{2}{3} k_a K_{E3} \sin 3\delta \tag{6}$$

the desired commutation advancing angle δ can be realized. Furthermore, $V_{\rm ref}$ is independent of the motor speed ω; hence, it can be set individually. With proper commutation advancing, the phase current can be kept in phase with the back EMF; thus, the equivalent power factor and electromagnetic torque will be maximized. Moreover, flux-weakening control can also be realized with this method.

It should be pointed out that if the commutation advancing angle is $\delta > 30$ electrical degrees, the EMF zero crossings will not appear in the terminal voltage waveforms; hence, all conventional zero-crossing-based sensorless strategies will be invalid [13]. However, the third harmonic EMF is always detectable [10]; thus, the proposed control strategy is still workable.

E. Access to Winding Neutral Point

In the proposed sensorless control, either with the ASIC or discrete components, access to the motor winding neutral point is required. As a disadvantage, this will complicate the motor structure and increase the cost. However, the neutral line is used for extracting the signal of third harmonic back EMF but not for carrying the power current. Therefore, its cable can be very thin.

On the other hand, it is reported in [9] that the third harmonic EMF can be extracted between the star point of the Y-connected

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resistor network and the dc bus midpoint; thus, access to the winding neutral point is not required. However, the resultant signal is significantly distorted by the freewheel diode conduction [10]; hence, this method is not suitable for the related sensorless control.

VI. Conclusion

Original ASIC ML4425 uses terminal voltages for sensorless control of BLDC motor drives. However, the motor commutation is significantly retarded during the ultrahigh-speed operation because the terminal voltage signals are distorted by the freewheel diode conduction. In this paper, an improved ASIC-based controller using the third harmonic back EMF instead of the terminal voltages has been developed. Its principle and implementation are presented, and its enhanced performance is verified with experimental results. Moreover, a novel controller with discrete components is also proposed, not only having the same features as the improved ASIC-based controller but also realizing controllable advanced commutation. Therefore, the current phase delay due to the winding inductance influence can be eliminated; hence, the system performance would be further improved.

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